

Appendix B

Estimation procedure

Due to the computational complexity of jointly estimating dynamic copula parameters and time-varying group assignments, a three-stage estimation procedure is adopted. In the first stage, the initial cluster assignments and copula parameters are estimated following [Oh & Patton \(2023\)](#). Static group assignments $\hat{\Gamma}_1$ are obtained by applying k -means clustering to the full sample using a misspecified static Gaussian copula. Conditional on these initial clusters, the copula parameters $\psi = [\omega_1^M, \dots, \omega_G^M, \omega_1^C, \dots, \omega_G^C, \alpha^M, \beta^M, \alpha^C, \beta^C, \nu, \zeta]'$ are estimated by maximizing the log-likelihood of the skewed t copula:

$$\hat{\psi} = \arg \max_{\psi} \sum_{t=1}^T \log \mathbf{c}_{\text{Skew } t, t}(\mathbf{u}_t; \psi \mid \hat{\Gamma}_1). \quad (9)$$

The skewed t copula nests both the Gaussian and symmetric t copulas as special cases. In both the simulation study and the empirical analysis, all three copulas are estimated for comparison.

In the second stage, time-varying cluster assignments are estimated. For each time period t , the filtered probabilities $\tau_{ig,t|t}$ are updated using the forward algorithm in Eqs. (4)-(5), again based on the misspecified static Gaussian copula. Each stock is then assigned to the cluster with the highest posterior probability, yielding the current group assignment $\hat{\Gamma}_t$.

In the third stage, the dynamic factors loadings are updated using the Generalized Autoregressive Score dynamics, based on the copula parameters obtained in the first stage. GAS adjusts the loadings in response to new information by leveraging the gradient of the log-likelihood function of the conditional copula. The update equations take the form:

$$\lambda_{g,t+1}^M = \omega_g^M + \alpha^M \frac{\partial \log \mathbf{c}_{\text{Skew } t, t}(\mathbf{x}_t; \mathbf{R}_t, \nu, \zeta)}{\partial \lambda_{g,t}^M} + \beta^M \lambda_{g,t}^M, \quad \text{for } g = 1, \dots, G \quad (10)$$

$$\lambda_{g,t+1}^C = \omega_g^C + \alpha^C \frac{\partial \log \mathbf{c}_{\text{Skew } t, t}(\mathbf{x}_t; \mathbf{R}_t, \nu, \zeta)}{\partial \lambda_{g,t}^C} + \beta^C \lambda_{g,t}^C, \quad \text{for } g = 1, \dots, G \quad (11)$$

where $\mathbf{x}_t = T_{\text{skew}}^{-1}(\mathbf{u}_t; \nu, \zeta)$, and $\mathbf{c}_{\text{Skew } t, t}(\mathbf{x}_t; \mathbf{R}_t, \nu, \zeta)$ denotes the conditional skewed t copula density. Finally, given the current factor loadings and group assignments, the log-likelihood of each observation is computed.

With respect to the remaining parameters, the number of clusters G must be chosen. Although the Akaike Information Criterion (AIC) is commonly used, it may overestimate the number of clusters ([Frühwirth-Schnatter, 2011](#)). Therefore, the optimal number of clusters is also validated based on out-of-sample forecasting performance. In theory, the number of clusters could vary over time if all firms were to transition out of a given cluster, implying $G = G_{\max}$. However, to preview the results, this has not occurred in practice, likely because the initial static clustering provides a sufficiently accurate starting point. Moreover, maintaining a fixed number of clusters over time is common in literature and appears to be a reasonable assumption ([Frühwirth-Schnatter, 2011](#); [João et al., 2023](#)). The decay parameter δ in Eq. (1), which controls the dynamics of cluster transitions, is selected by evaluating model performance over a grid of values.